Appendix C: IEEE 830 Template

Software Requirements Specification For E-shop

Version v4

Prepared by Srdjan Saranovic

04.03.2026.

Table of Contents

1. Introduction *4*

1.2. Document Conventions 4

1.3. Intended Audience and Reading Suggestions 4

1.4. Product Scope 5

1.5. References 5

2. Overall Description *5*

2.1. Product Perspective 5

2.2. Product Features 5

2.3. User Classes and Characteristics 6

2.4. Operating Environment 6

2.5. Design and Implementation Constraints 6

2.6. User Documentation 6

2.7. Assumptions and Dependencies 6

3. System Features *7*

3.1. Shopping Cart Management 7

3.2. Product Catalog Browsing 7

3.3. Secure Checkout and Payment 7

3.4. Order Management and Tracking 7

3.5. Inventory Management (Admin) 7

4. External Interface Requirements *7*

4.1. User Interfaces Overview 7

4.2. Hardware Interfaces 8

4.3. Software Interfaces 8

5. System Features/Modules *8*

5.1. Shopping Cart Module 8

5.1.1 Description and Priority 8

5.1.2 Stimulus/Response Sequences 8

5.1.3 Functional Requirements 8

6. Nonfunctional Requirements *9*

6.1. Performance 9

6.2. Security 9

7. Sequence/Activity and State Diagrams *9*

7.1 Sequence diagram 9

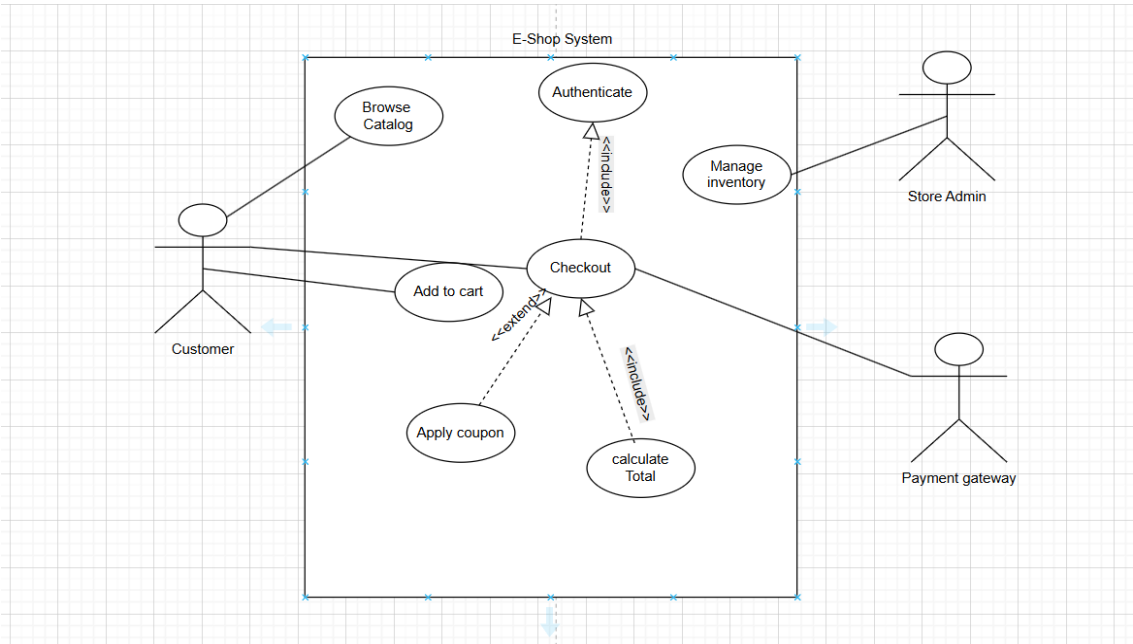
7.2 Activity diagram 10

7.3. State diagram 11

Revision History

Name	Date	Reason for Changes	Version
Srdjan Saranovic	04.03.2026	Initial version	V1
Srdjan Saranovic	24.03.2026.	Added UML Use Case Diagram and completed section 1	V1.1
Srdjan Saranovic	30.03.2026.	Added UML Class diagram and complete section 2 and 3	V2
Srdjan Saranovic	14.04.2026.	Added UML Sequence, Activity and State diagram, complete section 4 and 5	V3
Srdjan Saranovic	28.04.2026.	Complete diagrams describe, add table of contents and final review	V4

1. Introduction



1.1. Purpose

This Software Requirements Specification (SRS) defines the requirements for the E-shop System v1.0. It describes both functional and non-functional requirements for customer and admin interactions. The system enables users to browse products, add items to a shopping cart, and complete purchases through a checkout process, as illustrated in the use case diagram.

1.2. Document Conventions

This document follows standard SRS structure and uses consistent formatting:

- Functional requirements are labeled as **REQ-x.x**
- Non-functional requirements are labeled as **NF-x.x**
- Priorities are indicated (e.g., High, Medium)
- Use case names (e.g., *Checkout*, *Add to Cart*) match those shown in the diagram
  All requirements are written clearly and each has its own priority where applicable.

1.3. Intended Audience and Reading Suggestions

This document is intended for:

- **Developers** – to understand system functionality and implementation requirements
- **Testers** – to validate system behavior against requirements
- **Project managers** – to track scope and priorities
- **Designers** – to understand system interactions
- **Stakeholders** – to get an overview of system capabilities

It is recommended to first read the overview (Section 1), then the functional requirements (Section 5), followed by non-functional requirements (Section 6). The use case diagram should be reviewed alongside these sections for better understanding.

1.4. Product Scope

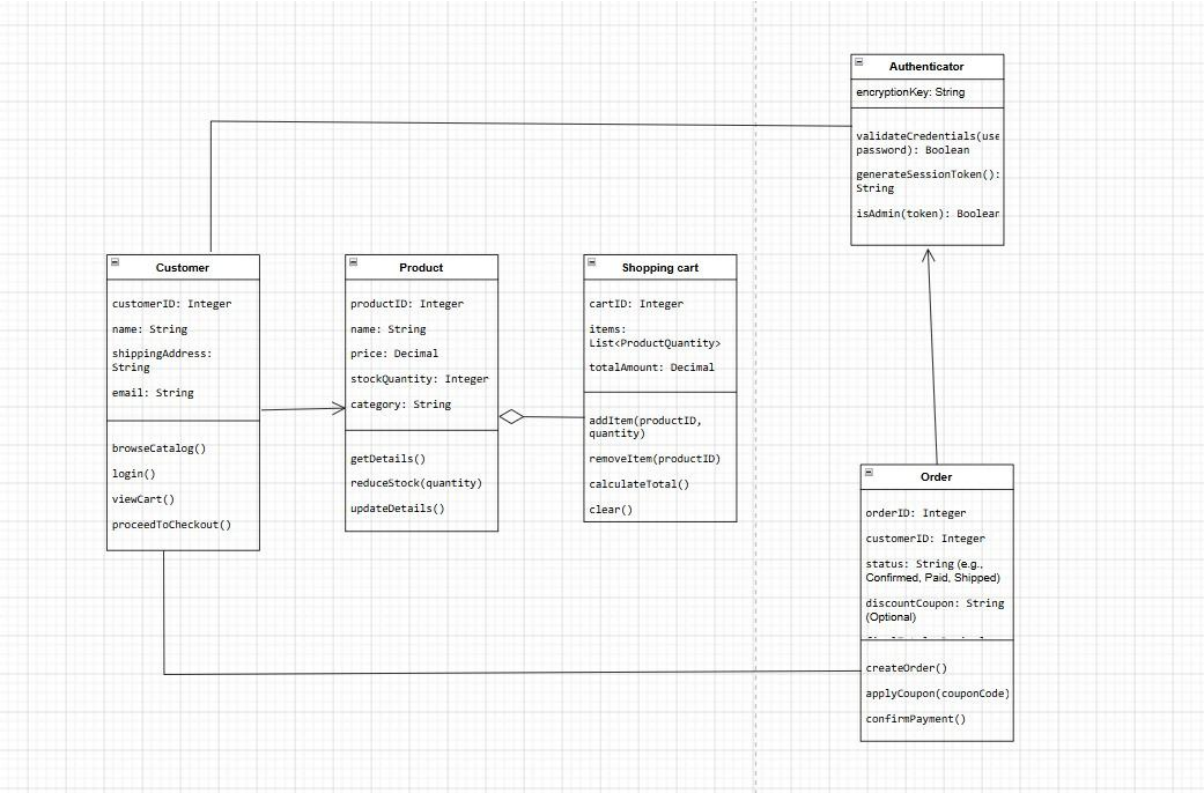
The software is an online retail platform that allows users to browse products, manage a shopping cart, and perform secure checkouts. The goal is to increase sales efficiency and provide a 24/7 shopping experience.

1.5. References

- IEEE 830 Software Requirements Specification Standard
- Course materials and assignment description
- UML Use Case Diagram guidelines
- BCrypt password hashing documentation
- SSL/TLS security standards documentation

2. Overall Description

2.1. Product Perspective



The class diagram shows the internal structure of the E-shop system. It includes main classes such as Customer, Product, Shopping Cart, Order, and Authenticator, along with their attributes and methods. The diagram also presents relationships between these classes and how they interact within the system.

2.2. Product Features

- Product Catalog Browsing.
- Shopping Cart Management.
- Secure Payment Processing.
- Order Tracking.

2.3. User Classes and Characteristics

The system has the following user classes:

- **Customer** – The main user of the system. Customers can browse products, add items to the shopping cart, and complete purchases. They have basic technical knowledge.
- **Store Admin** – Responsible for managing inventory and product information. Admin users have higher privileges and more system access.
- **Payment Gateway (External System)** – Handles payment processing. It is not a direct user but an external system interacting with the application.

The most important user class is the **Customer**, as the system is primarily designed for online shopping.

2.4. Operating Environment

The system is a web-based application that runs in a standard browser environment.

- **Platform:** Web application
- **Operating Systems:** Windows, macOS, Linux
- **Browsers:** Google Chrome, Mozilla Firefox, Safari, Edge
- **Server-side:** Application server with database support
- **Database:** Relational database (e.g., MySQL or PostgreSQL)

2.5. Design and Implementation Constraints

- The system must use **SSL/TLS encryption** for secure communication
- User passwords must be stored using **BCrypt hashing**
- The system must integrate with an external **payment gateway**
- The application should follow standard **web development practices**
- The database must support storing product, user, and order data

2.6. User Documentation

The system will include the following documentation:

- User guide for customers (how to browse, add to cart, checkout)
- Admin guide (how to manage inventory)
- Online help within the application
- Basic usage instructions on the website

2.7. Assumptions and Dependencies

- Users have internet access and a modern web browser
- The payment gateway service is available and functioning correctly
- The database system is properly configured and maintained
- Users have basic knowledge of using web applications
- The system depends on external services such as payment processing

3. System Features

This section provides a brief overview of the main system features. Detailed descriptions of these features are provided in Section 5.

3.1. Shopping Cart Management

This feature allows users to add products to the shopping cart, remove items, and update quantities. It supports the core shopping functionality of the system and is directly connected to the checkout process.

3.2. Product Catalog Browsing

This feature enables users to browse available products, view product details, and search through the catalog. It is the starting point for user interaction with the system.

3.3. Secure Checkout and Payment

This feature allows users to complete purchases by proceeding to checkout, authenticating themselves, and making secure payments through a payment gateway.

3.4. Order Management and Tracking

This feature enables users to place orders and track their status (e.g., confirmed, paid, shipped). It ensures users can monitor their purchases after checkout.

3.5. Inventory Management (Admin)

This feature allows the store administrator to manage product inventory, update product details, and control stock availability.

4. External Interface Requirements

4.1. User Interfaces Overview

The system provides a **graphical user interface (GUI)** accessible through a web browser. Users interact with the system using standard actions such as clicking buttons, selecting options, and entering text.

Customers can browse products, add items to the shopping cart, and proceed to checkout. During checkout, users receive feedback such as updated cart totals, confirmation messages, and error notifications (e.g., item out of stock).

The interface includes:

- Product catalog pages
- Shopping cart view
- Checkout page
- Order confirmation screen

The interface follows standard web design practices with clear navigation, consistent layout, and user-friendly forms. Error messages are displayed clearly, and important actions (e.g., payment confirmation) provide visual feedback.

4.2. Hardware Interfaces

The system does not require specific hardware interfaces. It runs on standard user devices such as desktop computers, laptops, tablets, and smartphones.

The application communicates with servers over the internet using standard networking protocols (HTTP/HTTPS). No direct hardware interaction is required.

4.3. Software Interfaces

The system interacts with the following software components:

- **Database system (e.g., MySQL, PostgreSQL) -** Stores user data, product information, and orders
- **Payment Gateway** - External service used to process payments securely
- **Web Browser** - Interface through which users access the system

The system sends and receives data such as:

- Product details
- User information
- Order and payment data

Communication between components is handled via standard web technologies and secure protocols (HTTPS).

5. System Features/Modules

This section describes the main system features in detail. Each feature includes its description, priority, stimulus/response sequences, and functional requirements.

5.1. Shopping Cart Module

This module represents the core functionality of the system and allows users to manage items in their shopping cart and proceed to checkout.

5.1.1 Description and Priority

Allows users to add, remove, and update quantities of items. **Priority: High**.

5.1.2 Stimulus/Response Sequences

Condition: User clicks "Add to Cart".

Response: System updates the cart count and saves the item session.

5.1.3 Functional Requirements

REQ-1.1: Upon clicking "Add to Cart", the system shall verify if the item is in stock.

REQ-1.2: The system shall calculate the total price including taxes at the checkout stage.

6. Nonfunctional Requirements

6.1. Performance

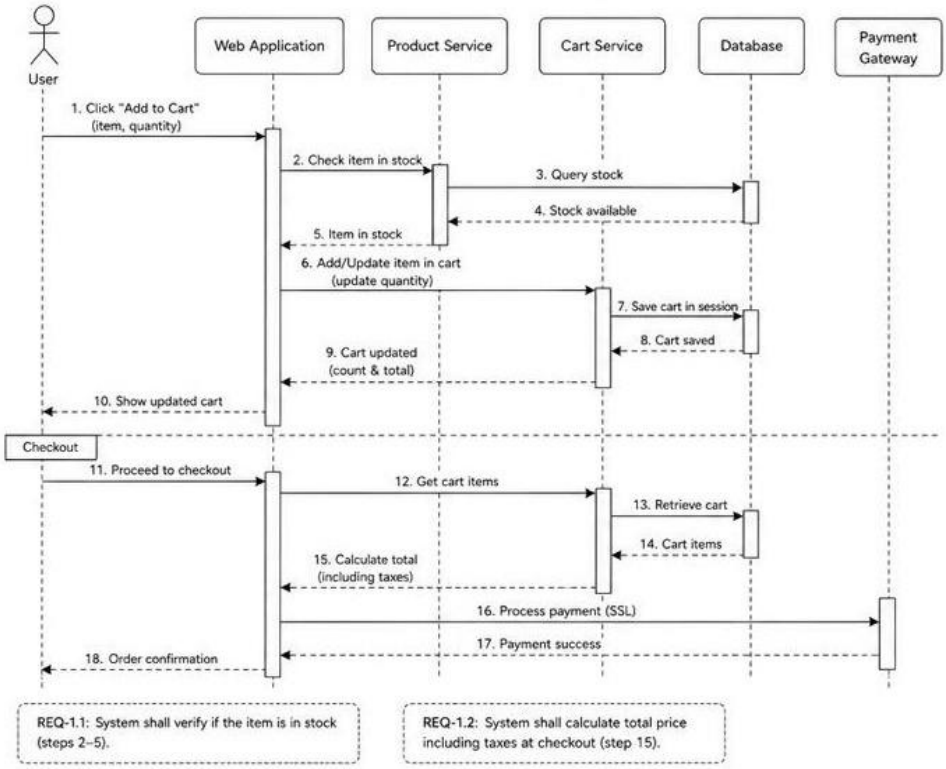
NF-1.1: The search results shall be displayed within 1.5 seconds.

6.2. Security

NF-2.1: The system shall use SSL for all payment transactions.
NF-2.2: User passwords must be hashed using BCrypt before being saved in the database.

7. Sequence/Activity and State Diagrams

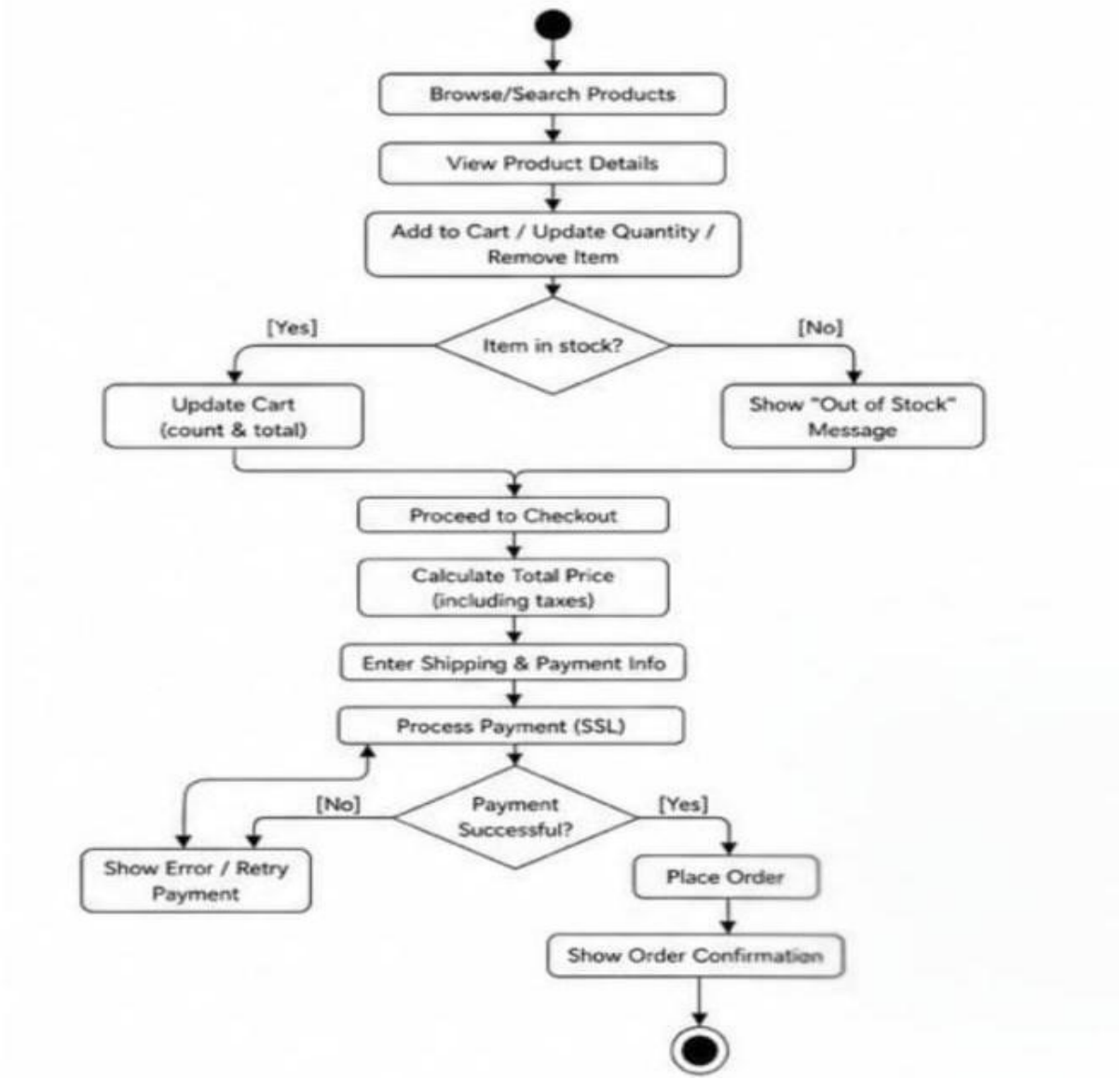
7.1 Sequence diagram



This sequence diagram shows how the E-shop system works during the process of adding items to the cart and completing a purchase. The interaction starts when the user clicks "Add to Cart" in the web application. The system then checks if the selected product is in stock by communicating with the product service and database.
If the item is available, it is added or updated in the shopping cart and saved in the session. After that, the system returns the updated cart with the total number of items and price.

When the user proceeds to checkout, the system retrieves the cart data and calculates the total price including taxes. This step matches the functional requirement for calculating the total at checkout. The system then processes the payment through the payment gateway using secure communication (SSL). If the payment is successful, an order confirmation is returned to the user. Overall, the diagram shows the interaction between system components and supports the main functionalities described in the SRS.

7.2 Activity diagram

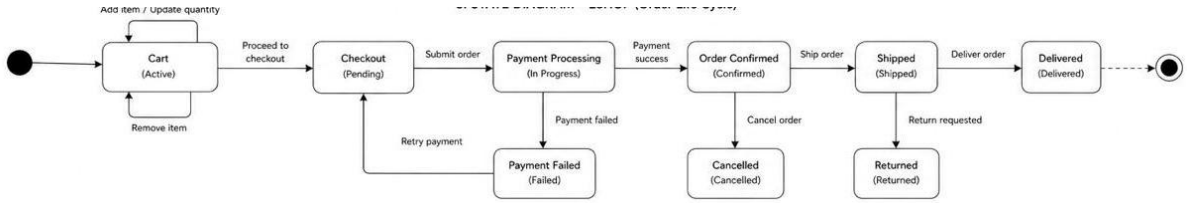


This activity diagram shows the workflow of the E-shop system from browsing products to completing an order. The process starts when the user browses or searches for products and views product details. The user can then add items to the cart, update quantities, or remove items. The system checks whether the selected item is in stock, which matches the requirement for stock verification.

If the item is available, the cart is updated with the total number of items and price. If not, the system displays an "Out of Stock" message. The user can then proceed to checkout, where the system calculates the total price including taxes. After that, the user enters shipping and payment information.

The system processes the payment using secure communication (SSL). If the payment is successful, the order is placed and a confirmation message is shown. If the payment fails, the system allows the user to retry the payment. The diagram represents the main flow and alternative paths in the shopping process.

7.3. State diagram



This state diagram shows the different states of an order in the E-shop system and how it changes during the process. The process starts in the "Cart" state, where the user can add or remove items. When the user proceeds to checkout, the state changes to "Checkout", and then to "Payment Processing" after submitting the order.

If the payment is successful, the order moves to the "Order Confirmed" state, and then continues to "Shipped" and finally "Delivered". If the payment fails, the system moves to the "Payment Failed" state, and the user can retry the payment and return to checkout.

The diagram also shows alternative scenarios such as cancelling the order or returning it after delivery. These actions move the order to "Cancelled" or "Returned" states. Overall, the diagram represents the lifecycle of an order and supports the order tracking functionality described in the system.